ECON 0100 | HW C2 | Taxes + Subsidies

Solutions

Answers and work are shown in the red blocks.

Due: Sunday, Oct. 18

Homework is designed to both test your knowledge and challenge you to apply familiar concepts in new applications. You are welcomed and encouraged to work in groups as long as your work is your own. Submit your answers on Gradescope when you're finished: each answer there is a selection.

Q1 | The Butterbeer Tax

The Ministry of Magic was in some financial trouble after their considerable expenditures during the war. They decided to impose a tax of 6 galleons on the sale of butterbeer as a source of funding. The supply and demand curves for butterbeer can be represented by the following equations.

$$D: P_B = 100 - \frac{1}{2}Q_D \qquad S: P_S = 20 + \frac{1}{2}Q_S$$

Use a graph and algebra to illustrate what happened to the market before and after the tax. Start with the pre-tax equilibrium. Then use the tax equation ($P_B = \tau + P_S$) to solve for equilibrium quantity and the two prices. Conclude with the remaining relevant components.

Pre-Tax Price ($\hat{P}$):	_____	Pre-Tax Quantity ($\hat{Q}$):	_____
Post-Tax Quantity (Q^τ):	_____		
Buyer Price (P_B):	_____	Seller Price (P_S):	_____
Government Revenue:	_____	Deadweight Loss (DWL):	_____
Tax Incidence on Buyers:	_____	Tax Incidence on Sellers:	_____

Pre-Tax Price: 60 galleons. Pre-Tax Quantity: 80 bottles. Post-Tax Quantity: 74 bottles. Buyer Price: 63 galleons. Seller Price: 57 galleons. Government Revenue: 444 galleons. Deadweight Loss: 18 galleons. Tax Incidence on Buyers: 3. Tax Incidence on Sellers: 3.

Without the tax, buyers and sellers face one price:

$$100 - \tfrac{1}{2}Q = 20 + \tfrac{1}{2}Q \quad \Rightarrow \quad Q = 80, \quad P = 60$$

Before the tax, consumer surplus is the triangle above 60, $\frac{1}{2} \times 80 \times 40 = 1600$, and producer surplus the triangle below it, also $\frac{1}{2} \times 80 \times 40 = 1600$.

With the tax, the buyer price sits 6 above the seller price, and the market still clears at one quantity. Slide the 6-galleon wedge in from the left until it gets stuck between the curves:

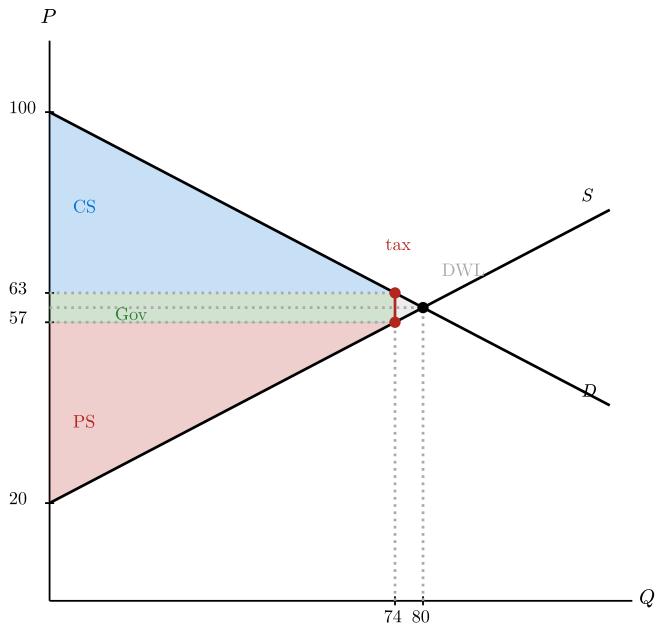


Figure 1: Tax equilibrium

$$P_B = 6 + P_S \quad \Rightarrow \quad 100 - \tfrac{1}{2}Q = 6 + (20 + \tfrac{1}{2}Q) \quad \Rightarrow \quad Q = 74$$

so $P_B = 100 - 37 = 63$ and $P_S = 20 + 37 = 57$.

Incidence: buyers pay $63 - 60 = 3$ more than before and sellers receive $60 - 57 = 3$ less. The two sides bear the tax equally because the curves are equally steep: neither side is better able to walk away than the other.

Welfare: consumer surplus shrinks from the triangle above 60 to the triangle above 63, $\frac{1}{2} \times 74 \times 37 = 1369$; producer surplus from the triangle below 60 to the one below 57, also 1369. The rectangle between the two prices over the 74 bottles still sold is government revenue, $6 \times 74 = 444$. The triangle between the curves over the 6 bottles no longer sold is deadweight loss:

$$\text{DWL} = \tfrac{1}{2} \times 6 \times (80 - 74) = 18$$

Check: $1600 + 1600 = 1369 + 1369 + 444 + 18$.

Q2 | The Butterbeer Subsidy

Now suppose that instead of the tax, the Ministry of Magic pays a 6 galleon subsidy on every butterbeer sold. Use a graph and algebra to discuss the impact of the subsidy on the market. Use the subsidy equation ($P_S = s + P_B$) to solve for equilibrium quantity, the two prices, and DWL.

Post-Subs Quantity (Q^{Sub}): _____ Deadweight Loss: _____

Buyer Price (P_B): _____ Seller Price (P_S): _____

For the bottles sold beyond the pre-subsidy quantity, how does marginal benefit compare with marginal cost?

a) Marginal benefit is above marginal cost, so these added trades raise total surplus.

b) Marginal benefit is below marginal cost, so these added trades reduce total surplus once the Ministry's spending is counted.

c) Marginal benefit equals marginal cost, so these added trades leave total surplus unchanged.

d) The subsidy lowers the sellers' marginal cost, so these added trades cost society nothing.

Post-Subs Quantity: 86 bottles. Deadweight Loss: 18 galleons. Buyer Price: 57 galleons. Seller Price: 63 galleons. The choice is **b)**.

A subsidy is the tax wedge turned around: the seller now receives 6 more than the buyer pays, so the wedge sits to the *right* of the no-policy equilibrium:

$$P_S = 6 + P_B \quad \Rightarrow \quad 20 + \tfrac{1}{2}Q = 6 + (100 - \tfrac{1}{2}Q) \quad \Rightarrow \quad Q =$$

so $P_B = 100 - 43 = 57$ and $P_S = 20 + 43 = 63$.

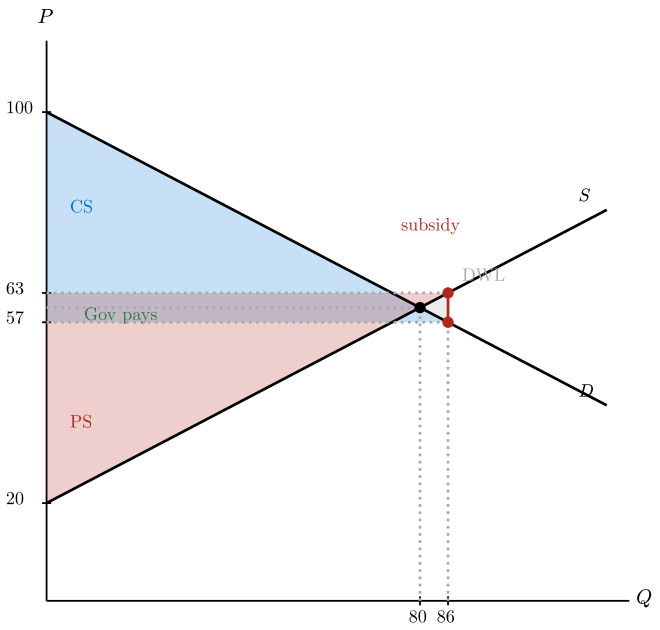


Figure 2: Subsidy equilibrium

Both sides are better off: buyers pay 3 less and sellers receive 3 more, so consumer surplus rises from 1600 to $\frac{1}{2} \times 86 \times 43 = 1849$ and producer surplus from 1600 to 1849. But the Ministry pays $6 \times 86 = 516$, more than the $249 + 249 = 498$ the buyers and sellers gain. The difference is the deadweight loss, the triangle between the curves over the six added bottles:

$$\text{DWL} = \tfrac{1}{2} \times 6 \times (86 - 80) = 18$$

b) Beyond the pre-subsidy 80 bottles, supply (marginal cost) is above demand (marginal benefit). The 81st through 86th bottles cost 60.5 to 63 galleons to make, but buyers value them at only 59.5 down to 57; they are made only because the Ministry pays the gap. Counting the Ministry's spending, each one lowers total surplus, by the 18 above. Not a): that is the bottles *before* 80. Not c): marginal benefit equals marginal cost only at 80. Not d): the subsidy changes who pays the cost, not the cost itself; the Ministry's galleons are a real cost to society.